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A phantom study of a protective trolley for neonatal radiographic imaging: new equipment to protect the operator from scatter radiation

Xinhong Wang, Mengxi Xu, Cong Chen, Zhongkun Bao, Hongzhu Wang, Jizhou Zhang, Zhen Wang, [Haipeng Liu](#)

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Research output: Contribution to journal › Article › peer-review

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Abstract

Chest radiography is commonly performed as a diagnostic tool of neonatal diseases. Contact-based radiation personal protective equipment (RPPE) has been widely used for radiation protection, but it does not provide full body protection and it is often shared between users, which has become a major

concern during the coronavirus disease 2019 (COVID-19) pandemic. To address these issues, we developed a novel trolley to protect radiographers against X-ray radiation by reducing scatter radiation during neonatal radiographic examinations. We measured the scatter radiation doses from a standard neonatal chest radiograph to the radiosensitive organs using a phantom operator in three protection scenarios (trolley, radiation personal protective equipment [RPPE], no protection) and at three distances. The results showed that the scatter radiation surface doses were significantly reduced when using the trolley compared with RPPE and with no protection at a short distance ($P < 0.05$ for both scenarios in all radiosensitive organs). The novel protective trolley provides a non-contact protective tool for radiographers against the hazard of scatter radiation during neonatal radiography examinations.

Original language English

Pages (from-to) 332-336

Number of pages 5

Journal [Pediatric Radiology](#)

Volume 53

Issue number 2

Early online date 26 Aug 2022

DOIs • <https://doi.org/10.1007/s00247-022-05488-3>

Publication status Published - Feb 2023

Funder

Funding Information: This study was supported by the National Natural Science Foundation of China (Grant No. 8170110234) and the National Key Research and Development Program of China (Grant No. 2018YFE0198400).

Keywords

- Chest radiography
- Neonatal intensive care unit
- Neonates
- Personal protective equipment
- Radiation protection
- Scatter radiation
- Trolley

ASJC Scopus subject areas

- Pediatrics, Perinatology, and Child Health
- Radiology Nuclear Medicine and imaging

Access to Document

- [10.1007/s00247-022-05488-3](https://doi.org/10.1007/s00247-022-05488-3)

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Wang, X., Xu, M., Chen, C., Bao, Z., Wang, H., Zhang, J., Wang, Z., & Liu, H. (2023). [A phantom study of a protective trolley for neonatal radiographic imaging: new equipment to protect the operator from scatter radiation.](#) **Pediatric Radiology**, 53(2), 332-336. <https://doi.org/10.1007/s00247-022-05488-3>

[A phantom study of a protective trolley for neonatal radiographic imaging: new equipment to protect the operator from scatter radiation.](#) / Wang, Xinhong; Xu, Mengxi; Chen, Cong et al.
In: [Pediatric Radiology](#), Vol. 53, No. 2, 02.2023, p. 332-336.

Research output: Contribution to journal › Article › peer-review

Wang, X, Xu, M, Chen, C, Bao, Z, Wang, H, Zhang, J, Wang, Z & Liu, H 2023, 'A phantom study of a protective trolley for neonatal radiographic imaging: new equipment to protect the operator from scatter radiation', **Pediatric Radiology**, vol. 53, no. 2, pp. 332-336. <https://doi.org/10.1007/s00247-022-05488-3>

Wang X, Xu M, Chen C, Bao Z, Wang H, Zhang J et al. [A phantom study of a protective trolley for neonatal radiographic imaging: new equipment to protect the operator from scatter radiation](#). *Pediatric Radiology*. 2023 Feb; 53(2):332-336. Epub 2022 Aug 26. doi: 10.1007/s00247-022-05488-3

Wang, Xinhong ; Xu, Mengxi ; Chen, Cong et al. / [A phantom study of a protective trolley for neonatal radiographic imaging: new equipment to protect the operator from scatter radiation](#). In: *Pediatric Radiology*. 2023 ; Vol. 53, No. 2. pp. 332-336.

@article{3000717d859740c4867d6c6493a8aa36,
 title = "A phantom study of a protective trolley for neonatal radiographic imaging: new equipment to protect the operator from scatter radiation",
 abstract = "Chest radiography is commonly performed as a diagnostic tool of neonatal diseases. Contact-based radiation personal protective equipment (RPPE) has been widely used for radiation protection, but it does not provide full body protection and it is often shared between users, which has become a major concern during the coronavirus disease 2019 (COVID-19) pandemic. To address these issues, we developed a novel trolley to protect radiographers against X-ray radiation by reducing scatter radiation during neonatal radiographic examinations. We measured the scatter radiation doses from a standard neonatal chest radiograph to the radiosensitive organs using a phantom operator in three protection scenarios (trolley, radiation personal protective equipment [RPPE], no protection) and at three distances. The results showed that the scatter radiation surface doses were significantly reduced when using the trolley compared with RPPE and with no protection at a short distance (P<0.05 for both scenarios in all radiosensitive organs). The novel protective trolley provides a non-contact protective tool for radiographers against the hazard of scatter radiation during neonatal radiography examinations.",
 keywords = "Chest radiography, Neonatal intensive care unit, Neonates, Personal protective equipment, Radiation protection, Scatter radiation, Trolley",
 author = "Xinhong Wang and Mengxi Xu and Cong Chen and Zhongkun Bao and Hongzhu Wang and Jizhou Zhang and Zhen Wang and Haipeng Liu",
 year = "2023",
 month = feb,
 doi = "10.1007/s00247-022-05488-3",
 language = "English",
 volume = "53",
 pages = "332--336",
 journal = "Pediatric Radiology",
 issn = "0301-0449",
 publisher = "Springer Science and Business Media LLC",
 number = "2",
}

TY - JOUR

T1 - A phantom study of a protective trolley for neonatal radiographic imaging: new equipment to protect the operator from scatter radiation

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PY - 2023/2

Y1 - 2023/2

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U2 - 10.1007/s00247-022-05488-3

DO - 10.1007/s00247-022-05488-3

M3 - Article

SN - 0301-0449

VL - 53

SP - 332

EP - 336

JO - Pediatric Radiology

JF - Pediatric Radiology

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